Task 4: Metric Selection, Imbalance Handling and Final Model Decision

1. Metric Selection

For this classification task, multiple evaluation metrics were used instead of relying only on accuracy. The dataset was evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC Curve, and AUC Score. Accuracy measures the overall percentage of correct predictions. While accuracy is useful, it may not fully represent model performance when class distributions are uneven.

The Confusion Matrix provides a detailed breakdown of True Positives, True Negatives, False Positives, and False Negatives. Precision measures how many predicted positive cases are actually positive, while Recall measures how many actual positive cases are correctly identified. F1-Score combines Precision and Recall into a single metric and is especially useful when both false positives and false negatives are important.

ROC (Receiver Operating Characteristic) Curve and AUC (Area Under the Curve) evaluate the model's ability to distinguish between classes across different classification thresholds. A higher AUC value indicates better class separation. For this project, AUC was considered an important metric because it provides a threshold-independent measure of performance.

2. Imbalance Handling

Class imbalance is a common problem in classification tasks where one class has significantly more observations than another. In such situations, a model can achieve high accuracy by favoring the majority class while performing poorly on the minority class. Therefore, additional metrics such as Precision, Recall, F1-Score, and AUC become more important.

To address potential imbalance, Logistic Regression was trained with the parameter **class_weight='balanced'**. This technique automatically adjusts class weights based on class frequencies and gives more importance to underrepresented classes. As a result, the model becomes more sensitive to minority-class predictions and reduces bias toward the majority class.

The balanced Logistic Regression model achieved an accuracy of approximately 95.61% while maintaining strong Precision and Recall. Although the overall accuracy was slightly lower than the standard Logistic Regression model, the balanced version improved fairness between classes and demonstrated better handling of potential class imbalance.

3. Final Model Decision

Three models were compared:

• Logistic Regression – Accuracy: 98.25%, AUC: 0.995
• Decision Tree Classifier – Accuracy: 93.86%, AUC: 0.934
• Logistic Regression (Balanced) – Accuracy: 95.61%

The standard Logistic Regression model produced the highest accuracy and the highest AUC score. Its confusion matrix showed only two misclassifications, indicating excellent predictive performance. Precision, Recall, and F1-Score were also very high for both classes, demonstrating strong classification capability.

The Decision Tree model performed reasonably well but showed lower accuracy and AUC compared to Logistic Regression. Decision Trees can be prone to overfitting and may not generalize as effectively without tuning.

The balanced Logistic Regression model successfully addressed class imbalance concerns and maintained strong classification results. However, its accuracy was slightly lower than the standard Logistic Regression model.

Based on the overall evaluation, **Logistic Regression was selected as the final model**. The decision was justified by its highest accuracy (98.25%), excellent AUC score (0.995), strong Precision and Recall values, and superior overall performance. The model demonstrated robust classification ability and the best balance between predictive power and reliability.